

**TIP:** Read the contents of the *WorldSeriesWinners.txt* file into an array, or an `ArrayList`. When the user enters the name of a team, the program should step through the array or `ArrayList`, counting the number of times the selected team appears.

### 17. 2D Array Operations

Write a program that creates a two-dimensional array initialized with test data. Use any primitive data type that you wish. The program should have the following methods:

- `getTotal`. This method should accept a two-dimensional array as its argument and return the total of all the values in the array.
- `getAverage`. This method should accept a two-dimensional array as its argument and return the average of all the values in the array.
- `getRowTotal`. This method should accept a two-dimensional array as its first argument and an integer as its second argument. The second argument should be the subscript of a row in the array. The method should return the total of the values in the specified row.
- `getColumnTotal`. This method should accept a two-dimensional array as its first argument and an integer as its second argument. The second argument should be the subscript of a column in the array. The method should return the total of the values in the specified column.
- `getHighestInRow`. This method should accept a two-dimensional array as its first argument and an integer as its second argument. The second argument should be the subscript of a row in the array. The method should return the highest value in the specified row of the array.
- `getLowestInRow`. This method should accept a two-dimensional array as its first argument and an integer as its second argument. The second argument should be the subscript of a row in the array. The method should return the lowest value in the specified row of the array.

Demonstrate each of the methods in this program.

### 18. Phone Book `ArrayList`

Write a class named `PhoneBookEntry` that has fields for a person's name and phone number. The class should have a constructor and appropriate accessor and mutator methods. Then write a program that creates at least five `PhoneBookEntry` objects and stores them in an `ArrayList`. Use a loop to display the contents of each object in the `ArrayList`.

### 19. Trivia Game

In this programming challenge, you will create a simple trivia game for two players. The program will work like this:

- Starting with player 1, each player gets a turn at answering 5 trivia questions. (There are 10 questions, 5 for each player.) When a question is displayed, four possible answers are also displayed. Only one of the answers is correct, and if the player selects the correct answer, he or she earns a point.
- After answers have been selected for all of the questions, the program displays the number of points earned by each player and declares the player with the highest number of points the winner.